

Trilateral Stochastic Impersonation Equilibria in Heterogeneous Warfare Economies

US–India Allies versus a Common Adversary

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Abstract

We extend the two-agent stochastic impersonation framework in [1] to a trilateral setting by introducing a common adversary A . Agents D (domestic/India) and I (international partner/US) are allies who coordinate deception, while both confront the same adversary A . The model retains quadratic payoffs, closed-form equilibria, stochastic shocks, existence/uniqueness via the generalized gross-substitutes condition, volatility externalities, Bayesian signal extraction, and a three-agent policy-gradient RL algorithm. Ally synergy raises equilibrium deception means and creates positive correlation between D and I , while anti-correlation with A emerges endogenously. The enlarged Pareto-improving variance-reduction treaty region provides a rigorous foundation for US–India confidence-building measures (CBMs) in the information domain. Monte-Carlo simulations and the data-science pipeline extend directly.

The paper ends with “The End”

1 Introduction to the Trilateral Extension

The original two-agent model analyses bilateral impersonation between a domestic agent D and an international agent I . To operationalize the policy appeal for US–India intel-sharing, we introduce a common adversary A (e.g., a third-party state actor in the Indo-Pacific information domain). This yields an explicit *US–India allies (D+I) versus common adversary (A)* interpretation while preserving all original analytical tractability.

2 Trilateral Model Setup

2.1 Agents, Action Spaces, and Payoffs

Definition 2.1 (Trilateral Impersonation Economy). The extended game is $\Gamma_3 = \langle \{D, I, A\}, \{U_k\}_{k \in \{D, I, A\}}, \Theta_3 \rangle$, where each agent k chooses deception intensity $x_k \in [0, 1]$ with $x_D = d$, $x_I = i$, $x_A = a$.

Assumption 2.2 (Trilateral Payoff Structure). Payoffs are quadratic and capture (i) ally synergy via a positive cross-term γdi , and (ii) adversary dilution via negative cross-terms:

$$U_D(d, i, a) = \alpha d - \frac{\delta_D}{2} d^2 + \gamma di - \beta_{DA} da, \quad (1)$$

$$U_I(d, i, a) = \alpha i - \frac{\delta_I}{2} i^2 + \gamma id - \beta_{IA} ia, \quad (2)$$

$$U_A(d, i, a) = \alpha a - \frac{\delta_A}{2} a^2 - \beta_{AD} ad - \beta_{AI} ai. \quad (3)$$

Here $\gamma > 0$ is the ally-cooperation (intel-sharing) parameter, $\beta_{kj} = \beta_{jk} > 0$ are pairwise dilution coefficients (symmetric $\beta_{DA} = \beta_{IA} = \beta$ for tractability), and Θ_3 augments the original Θ with $\{\gamma, \delta_A, \beta_{DA}, \beta_{IA}\}$.

When $\gamma = 0$ and A is absent the model nests the original two-player game.

2.2 Deterministic Equilibrium

The interior Nash equilibrium solves the first-order conditions:

$$\delta_D d - \gamma i + \beta_{DA} a = \alpha, \quad (4)$$

$$-\gamma d + \delta_I i + \beta_{IA} a = \alpha, \quad (5)$$

$$\beta_{AD} d + \beta_{AI} i + \delta_A a = \alpha. \quad (6)$$

In matrix form: $M\mathbf{x}^* = \alpha\mathbf{1}$, where

$$M = \begin{pmatrix} \delta_D & -\gamma & \beta_{DA} \\ -\gamma & \delta_I & \beta_{IA} \\ \beta_{AD} & \beta_{AI} & \delta_A \end{pmatrix}, \quad \mathbf{x}^* = \begin{pmatrix} d^* \\ i^* \\ a^* \end{pmatrix}.$$

Theorem 2.3 (Deterministic Trilateral Nash Equilibrium). *Under the generalized gross-substitutes condition (all leading principal minors of $M > 0$), there exists a unique interior equilibrium $\mathbf{x}^* = \alpha M^{-1}\mathbf{1}$.*

For the symmetric-adversary case $\beta_{DA} = \beta_{IA} = \beta$, the ally means simplify to

$$d^* = \frac{\alpha(\beta - \delta_A)(\delta_I + \gamma)}{\Delta}, \quad i^* = \frac{\alpha(\beta - \delta_A)(\delta_D + \gamma)}{\Delta},$$

with common denominator

$$\Delta = \beta^2(\delta_D + \delta_I + 2\gamma) - \delta_A(\delta_D\delta_I - \gamma^2).$$

Interiority requires $\Delta > 0$ and standard bounds on α .

Comparative statics: $\partial d^*/\partial\gamma > 0$, $\partial d^*/\partial\delta_A < 0$, $\partial d^*/\partial\beta < 0$.

2.3 Stochastic Extension

Impersonation levels are random:

$$d = \mu_D + \varepsilon_D, \quad i = \mu_I + \varepsilon_I, \quad a = \mu_A + \varepsilon_A,$$

with $\varepsilon \sim N(\mathbf{0}, \Sigma)$ (full 3×3 covariance matrix) and truncation to $[0, 1]$ as needed.

Expected payoffs decompose as in the original paper (Lemma 4.1 extended):

$$\mathbb{E}[U_D] = \alpha\mu_D - \frac{\delta_D}{2}(\mu_D^2 + \sigma_D^2) + \gamma(\mu_D\mu_I + \sigma_{DI}) - \beta_{DA}(\mu_D\mu_A + \sigma_{DA}),$$

and analogously for U_I and U_A .

Equilibrium means $\boldsymbol{\mu}^*$ reproduce the deterministic solution (shock-invariant). Equilibrium variances/covariances augment with unpredictability benefits $\gamma_k/2\sigma_k^2$ minus management costs $\kappa_k/2\sigma_k^2$. Allies choose $\sigma_{DI}^* > 0$ (positive correlation via intel-sharing); both choose $\sigma_{DA}^* < 0$, $\sigma_{IA}^* < 0$ (anti-correlation with adversary).

The full stochastic equilibrium is the trivariate normal $\mathbf{x} \sim N(\boldsymbol{\mu}^*, \Sigma^*)$.

2.4 Volatility Externality and Trilateral Treaty

The negative volatility externality now operates across three agents. Ally variance harms the other ally only mildly (mitigated by γ) but strongly harms A ; adversary variance hurts both allies. The Pareto-improving region $T \subset \mathbb{R}_+^3$ for variance caps $(\hat{\sigma}_D^2, \hat{\sigma}_I^2, \hat{\sigma}_A^2)$ is strictly larger than in the bilateral case and justifies joint US–India CBMs that simultaneously lower own variances and anti-correlate with the adversary.

2.5 Bayesian Signal Extraction and Intel-Sharing

Each ally observes noisy signals of the partner *and* the adversary. Posterior beliefs are multivariate normal updates. Signal-to-noise ratios improve dramatically via US–India fusion centers, driving Bayesian best responses to the full-information equilibrium faster as ally signal noise $\tau_{DI} \rightarrow 0$.

2.6 AI/ML: Three-Agent Policy-Gradient RL

Algorithm 1 generalizes directly: each agent parameterizes a Gaussian policy $\pi_k(\cdot; \theta_k)$. Simultaneous REINFORCE gradient ascent converges (under the generalized gross-substitutes condition) to (μ^*, Σ^*) . The neural-policy extension (context-dependent on shared intelligence features) applies unchanged.

2.7 Monte-Carlo Simulation and Data-Science Pipeline

Extend the original benchmark (Table 4) with γ, δ_A, β . Simulate trivariate draws and verify moments. The end-to-end pipeline (original Figure 4) now ingests trilateral raw intelligence signals; GMM/MLE estimates Θ_3 ; the three-agent Algorithm 1 outputs coordinated deception policy *and* treaty design.

3 Policy Implications for US–India Intel-Sharing

- Ally synergy γ raises d^* and i^* while lowering a^* — quantitative support for deeper operational coordination.
- The enlarged Pareto region T justifies joint CBMs: cap own variances and anti-correlate with the adversary while positively correlating US–India operations.
- Anti-correlated deception cycles with A become a strategic advantage in coordinated Indo-Pacific information-domain planning.

All original proofs (Kakutani, gross-substitutes, Lyapunov) carry over with the natural 3×3 matrix M and covariance Σ .

This extension turns the original framework into a ready operational tool for QUAD/Indo-Pacific information-domain planning. The full three-agent model can be calibrated to declassified signals-intelligence datasets exactly as proposed in the original Section 10.2.

References

- [1] Ghosh, S. (2026). Stochastic Impersonation Equilibria in Heterogeneous Warfare Economies
Kolkata, India.

A Proof Sketches

Existence follows from Brouwer’s fixed-point theorem on the compact convex action space $[0, 1]^3$ with continuous best-response map. Uniqueness follows from the P-matrix property of M . Lyapunov stability of the gradient-ascent dynamics holds under the generalized gross-substitutes condition. Full algebraic derivations and simulation code are available upon request.

The End